QP CODE: 25900218

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATIONS, MARCH 2025
Second Semester

DSC- CORE COURSE- MG2DSCZIM100, MG2DSCZGY100

ENVIRONMENTAL BIOLOGY
(2024 Admission onwards)

Duration: 1.5 Hours **Maximum Marks: 50**

Part A
Multiple Choice Questions
Answer A**ll** questions. Each question carries **1** mark

1. The pyramid of numbers can be: **[K] / [CO1]**

Ans. **Give full mark** if attempted any option

2. Which of the following human activities is directly responsible for increasing atmospheric carbon dioxide levels? **[K] / [CO1]**

Ans. d) All of the above.

3. Branch of science that deals with the study of population is called: **[K] / [CO2]**

Ans: a) Demography

4. Select the reason why biodiversity is important for ecosystem stability. **[U] / [CO3]**

Ans.a) Resilience

5. Identify the sampling method which involves placing a square frame at random locations. **[K] / [CO3]**

Ans: a) Quadrat

6. Identify which of the following is NOT an ecosystem service. **[K] / [CO3]**

Ans: b) Pollution

7. Floods can lead to outbreaks of: **[K] / [CO4]**

Ans: c) Cholera

8. Potassium Iodide pills were distributed to survivors of Chernobyl Nuclear Power Plant explosion to prevent: **[U] / [CO4]**

Ans: a) Thyroid Cancer

9. Kyoto Protocol was aimed at: **[K] / [CO4]**

Ans: a) Reduction of GHG emissions

10. Select how afforestation can help with carbon sequestration from the options below.

Ans: b) By capturing CO_2 through photosynthesis **[U] / [CO4]**

11. The primary objective of the Wildlife Protection Act of 1972 is: **[K] / [CO4]**

Ans: b) Safeguard the interests of wildlife

12. The Environment Protection (Amendment) Rules, 2022, focus on which of the following aspects regarding industrial activities? **[U] / [CO4]**

Ans: a) Strict hazardous waste management **[12 x 1 = 12 marks]**

Part B

Short Answer Questions

Answer any **Six** questions. Each question carries **3** marks

13. Describe the eco-friendly approaches that contribute to environmental conservation. **[U] / [CO1]**

Ans: Use of renewable energy (solar, wind), reducing consumption, adopting sustainable transportation, recycling and waste reduction, effective waste management, afforestation and reforestation, sustainable agriculture, water and energy conservation etc. (Description of **any three** or **only name** of any **six** eco-friendly approaches).

14. Justify why air is considered an important abiotic factor in an ecosystem. ***(Liberal Valuation)*** **[U] / [CO1]**

Ans: Explanation of air as an abiotic factor (1 mark). Air provides essential gases (oxygen for respiration, CO_2 for photosynthesis). Influences climate and weather patterns. Helps in seed dispersal and pollination (wind) - 2 mark for justification.

15. List the peculiarities of a Linear Survivorship Curve. ***(Liberal Valuation)*** **[K] / [CO2]**

Ans: Constant mortality rate at all ages. Characteristic of organisms with low parental care & moderate offspring number. Equal chance of dying at any age. Graph is a Straight line.

16. Compare intraspecific competition and interspecific competition. **[U] / [CO2]**

Ans: **Intraspecific:** Competition among individuals of the same species.
Interspecific: Competition between individuals of different species.
Both affect resource availability and population dynamics. (Definition-1 mark each; example. for each- 0.5 mark each/ the impact of competition- 0.5 mark).

17. Identify the key goals of the World Wide Fund for Nature (WWF). **[U] / [CO3]**

Ans: Conservation or Protection of biodiversity. Sustainable use of renewable natural resources. Reduction of pollution and wasteful consumption. Fair and equitable sharing of benefits **(Any three)**

18. Summarize the highlights of G20 Summit 2023. **[U] / [CO3]**

Ans: Global cooperation on climate change. Sustainable development goals (SDGs). Biodiversity conservation. Food security. Green energy initiatives. Women-led development. Climate and Sustainable Finance. Global Economic Recovery. (List **any six** or describe **any three**).

19. Discuss the impact of over exploitation on biodiversity. **[U] / [CO3]**

Ans: Population declines; species extinction and habitat destruction. Disruption of ecosystem balance- less resilient ecosystem. Loss of species thereby reducing genetic diversity/biodiversity. (Any 2 with explanation. 1.5 mark each).

20. Outline the primary goals of establishing biosphere reserves. (*Liberal Valuaution)* **[U] / [CO3]**

Ans: Conservation of ecosystems, species, and genetic resources. Promoting sustainable development. Providing areas for research and education.

[6x 3 = 18 marks]

Part C

Short Essay Questions

Answer any **Four** questions. Each question carries **5** marks

21. Energy flow in an ecosystem is unidirectional. Justify. **[U] / [CO1]**

Ans: Define energy flow. Energy flows from sun → producers → consumers → decomposers. Cannot be reused by the previous trophic level. Energy cannot be destroyed (1st Law) Loss of energy at each step as heat (Second Law of Thermodynamics). Ensures ecosystem stability.

(Define energy flow-1 mark. Explain energy flow- 1 mark. Thermodynamics principles- 2 mark, 1 example- 1 mark)

22. Explain the concept of community stratification in ecological systems. (*Liberal Valuaution)* **[U] / [CO2]**

Ans: Vertical layering of plants and animals.
Improves use of light, water, nutrients. Reduces direct competition.
Examples: Canopy, small trees and shrubs, shrub layer, forest floor with decomposers and small animals in forests (terrestrial ecosystems).
In aquatic (floating plants on top, submerged plants below, and bottom-dwelling creatures).

23. Describe the key legal provisions under the Biological Diversity Act, 2002. **[U] / [CO3]**

Ans: Conservation of biological diversity. Sustainable use of biological resources. Fair and equitable sharing of benefits. Establishment of National Biodiversity Authority (NBA).
Regulation of access to genetic resources. (Brief description of **any five)**.

24. Describe the origin and impact of the Chipko Movement. **[U] / [CO3]**

Ans: Started in 1973, former Uttar Pradesh portion. Issue- massive felling of tress as a consequence of urbanisation. Forests- source of livelihood for villagers, role in flood control/ landslides. Villagers (women) hugged trees- a non-violent act- to stop commercial deforestation. Led by Sunderlal Bahuguna. Increased forest conservation awareness and prevented deforestation. Influenced India's forest protection policies.

25. Identify the key policies and regulations related to watershed management in India. **[U] / [CO4]**

Ans: Focus on soil conservation, water harvesting, afforestation, land use management. Promotes community participation. Helps in improving rural livelihoods and agriculture.

Construction of check dams, percolation tanks, dug-out ponds. Multicropping to promote sustainability and control soil run-off. Government policies like mandatory water harvesting structures in every new building.

Drought Prone Areas Programme (DPAP), Desert Development Programme (DPP), Integrated wasteland Development Programme (IWDP), Haryali project, Neeru-Meeru, Arvani Pani Sansad. (Either the specific initiatives OR the strategies- 5)

26. Describe the different types of solid waste and their sources. **[U] / [CO4]**

Ans: Municipal waste (household garbage, plastics, paper). Industrial waste (chemical residues, metals, construction and demolition waste). Biomedical waste (hospital waste like syringes, dressings). E-waste (discarded electronics like phones, computers). Agricultural waste (crop residues, animal manure). Hazardous waste (radioactive wastes, inflammables, etc)- (Any **five** types and their sources).

[4x 5 = 20 marks]